SENTINEL-RISK

Unified AI/NLP Risk Engine & Strategic Wholesale Portfolio Stress Tester

S&P; Global & CRISIL Campus Hackathon

Candidate Name: [Candidate Full Name] | Track: Downstream Module B
College / Campus: [Candidate University / Institute]

Slide 2: Problem & Approach

The Core Business & Technical Challenge

- **Information Latency Gap:** Modern financial shocks (geopolitical blockades, banking liquidity freezes, surprise rate hikes) break first across unstructured text feeds (news & Twitter/X). Traditional banking risk architectures rely on lagging, end-of-day structured price batches, leaving capital reserves exposed.
- **Unstructured to Structured Hurdle:** Ingesting high-noise social streams and mapping subjective headlines into quantitative, machine-readable financial parameters requires domain-calibrated models.

Our Solution Approach

- **Unified AI/NLP Risk Engine:** Dual-source streaming ingestion emitting the required triad: Sentiment Score (-1.0 to +1.0), 7-category Event Classification, and Market Impact Score (1.0 to 10.0).
- **Downstream Module B (Stress Tester):** Automatic activation when Impact Score ≥ 7.0, applying calibrated multi-factor shocks to a $1.25B wholesale banking book across Loans, Bonds, Derivatives, and CLOs under Basel III standards.

Slide 3: System Design & Architecture

End-to-End Modular Data Pipeline

1. **Ingestion:** Multi-source pipeline combining Reuters/Bloomberg RSS and Twitter/X feeds with cashtag normalization ($JPM, $WTI).
2. **AI/NLP Risk Engine:** Sub-45ms inference combining Loughran-McDonald financial dictionaries, TextBlob polarity, and systemic severity scoring.
3. **Automated Stress Testing:** Event-driven trigger threshold routing high-severity shocks to wholesale banking asset revaluation models.
4. **Delivery Interfaces:** Enterprise FastAPI REST services with Swagger docs and Streamlit interactive executive terminal.

(Refer to docs/architecture.png for high-resolution visual architecture diagram)

Slide 4: Implementation Highlights & Tech Stack

Key Modules & Engineering Decisions

- **Financial Sentiment Analysis (src/nlp_engine):** Tailored for financial text where terms like 'liability', 'default', or 'rate hike' differ from general conversational polarity. Outputs continuous [-1.0, +1.0] score.
- **Event Classification:** Maps events into 7 standard institutional taxonomies: Geopolitical, Credit Event, Macroeconomic, Regulatory/Legal, Cyber Risk, M&A;, and Product Launch.
- **$1.25B Wholesale Banking Book:** 15 assets spanning CRE loans, project finance, leveraged loans, US Treasuries, subordinated debt, IRS, CDS, and CLO mezzanine tranches.
- **FastAPI Backend & Streamlit Cockpit:** Production Pydantic v2 schemas, interactive Plotly loss waterfalls, and a live jury interactive event simulator.

Slide 5: Key Results & Quantitative Demonstration

Performance Benchmarks & Scenario Simulation

Metric / Dimension	Traditional Risk Architecture	Sentinel-Risk Platform	Advantage
End-to-End Latency	Overnight / Monthly batch	< 45 milliseconds	> 1,000x Faster
Data Ingestion	Structured prices only	Unstructured News + Tweets	Early Warning Lead Time
Trigger Mechanism	Manual quarterly cycle	Automated Event Threshold (>=7.0)	Zero-Lag Response
Capital Integration	Separated ad-hoc models	Basel III Standardized RWA & CAR	Instant Breach Alert

Simulated Geopolitical Conflict Shock (Score: 9.1/10):

- Portfolio Drawdown: **-$82.60M (-7.55%)** | RWA Expansion: **+$189.2M**
- Capital Adequacy Ratio (CAR): Drops from **11.5% to 8.1%** (Flags Basel III Regulatory Capital Breach below 10.5%).

Slide 6: Domain Impact & Business Value

Transforming Wholesale Risk Management for S&P; Global & CRISIL

- **Proactive Capital Preservation:** Shifts wholesale treasury desks from reactive damage control to intraday dynamic hedging (purchasing CDS protection, adjusting credit line draws).
- **Automated Early Warning Signals:** Detects deteriorating borrower credit quality weeks before formal rating agency downgrades.
- **Regulatory Auditability:** Full transparency into mathematical shock transmission mechanisms satisfying Federal Reserve CCAR, ECB SREP, and RBI supervisory directives.
- **Client Value-Add:** Enables financial intelligence platforms to offer interactive scenario testing tools directly to institutional asset managers.

Slide 7: Limitations & Future Roadmap

Current Assumptions

- Synthetic book models 15 institutional asset archetypes ($1.25B); production scale involves hundreds of thousands of individual loan contracts.
- Factor shocks assume linear scaling by impact score; crisis non-linear copula correlations can be incorporated.

Production Roadmap

- **1. Quantized Local LLMs:** Fine-tuned FinBERT / Llama-3-Fin running on on-premise banking infrastructure for multi-language global news.
- **2. Apache Kafka Streaming:** Enterprise message bus for processing 100,000+ news articles and social posts per second.
- **3. Algorithmic Hedging Engine:** Automated recommendation of optimal derivative hedges to minimize stressed capital drawdowns.